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(54) CONNECTION STRUCTURE AND PARALLEL LINK ROBOT

(71) Applicant: FANUC CORPORATION, Yamanashi (JP)

(72) Inventor: Changmiao YU, Yamanashi (JP)

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(57) ABSTRACT

A connection structure includes bushings attached to two links so as to be rotatable about an axis perpendicular to a plane containing two longitudinal axes of the two links; a biasing mechanism spanning between the bushings of the two links and applying an elastic restoring force in a direction in which the two bushings move closer together; and a connection member having two attachment holes through which the bushings can pass in a direction of the axis and regulating a distance between the bushings to be less than or equal to a prescribed distance. The bushings include claw portions projecting radially outward. The attachment holes are formed in a shape allowing the bushings to pass therethrough at a prescribed attachment phase around the axis coinciding with the claw portions of the bushings. A phase of the bushings is regulated to a phase that does not match the attachment phase.

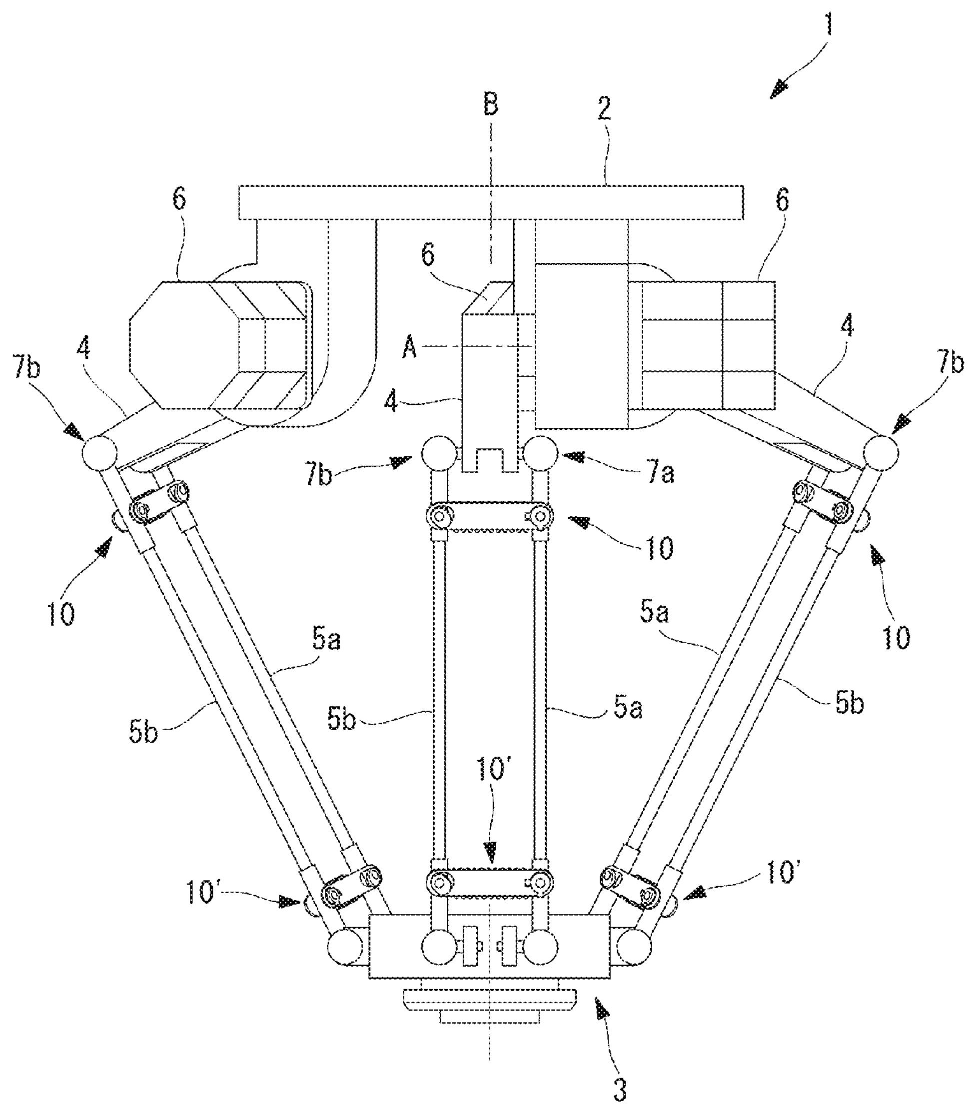


FIG. 1

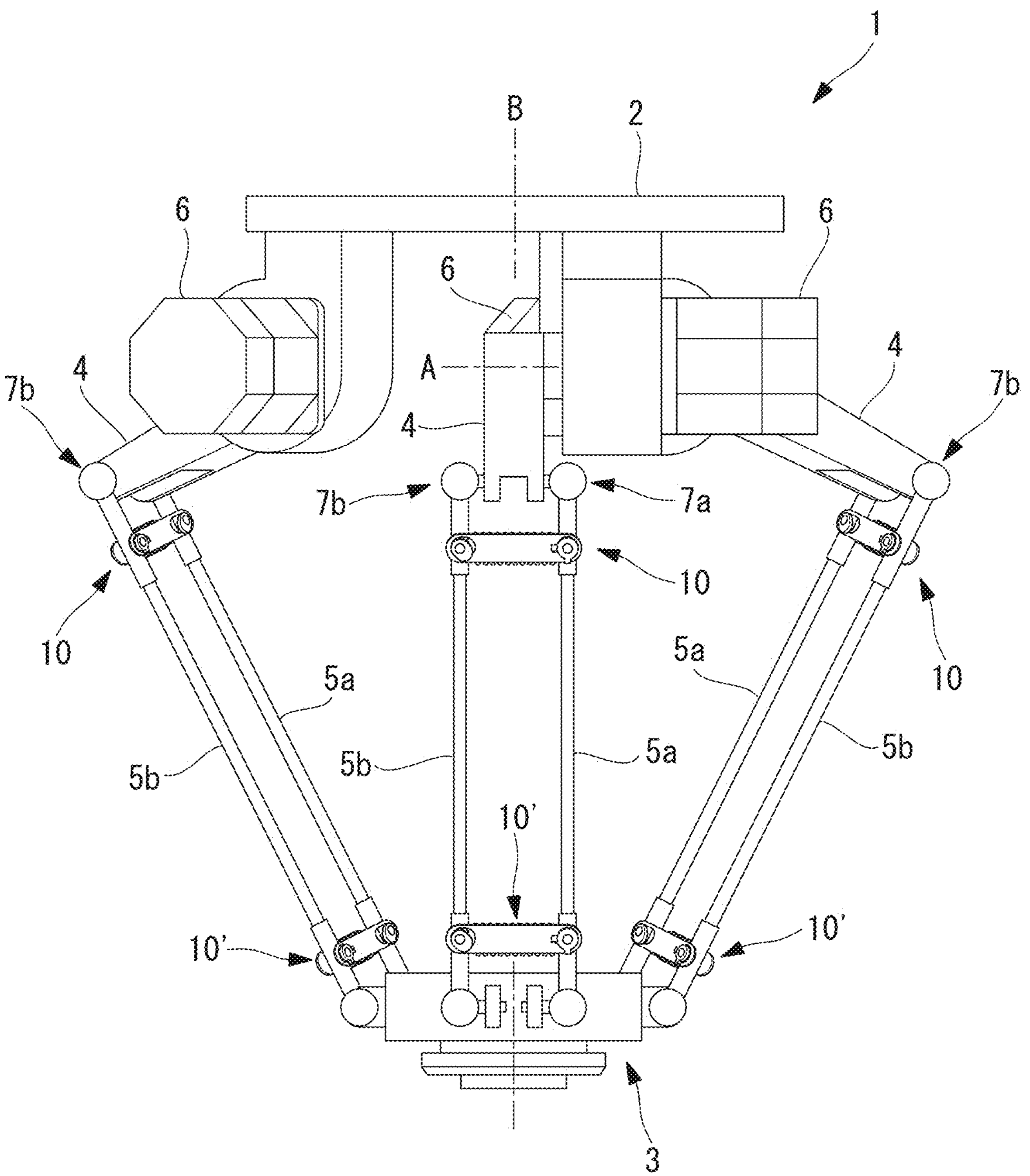


FIG. 2

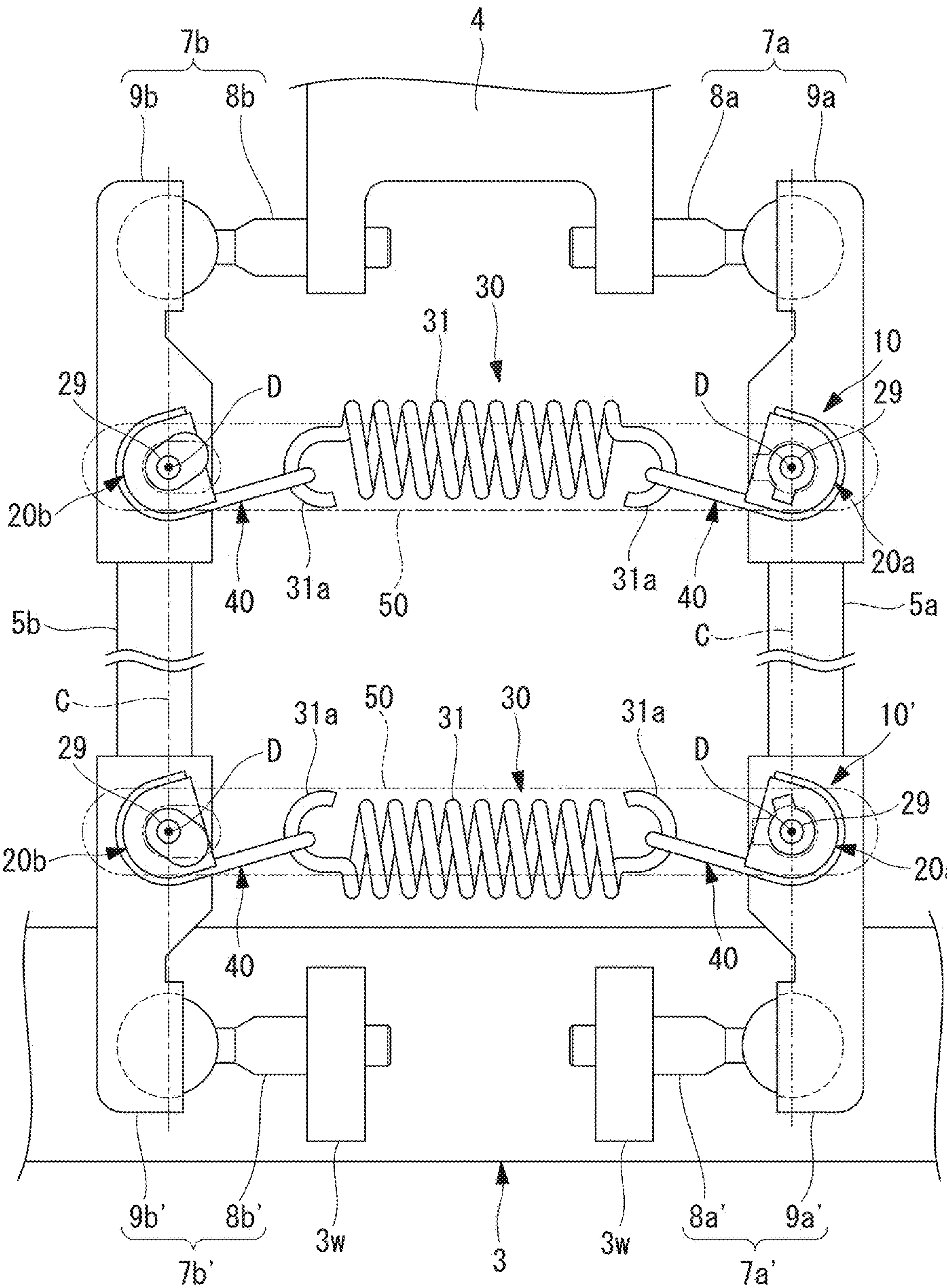

4
7b
7a
9b
8b
8a
9a
30
31
10
29
D
D
29
20b
40
31a
50
31a
40
20a
5b
5a
C
C
31a
50
31
30
31a
10'
29
D
D
29
20b
20a
40
40
9b'
8b'
3w
3w
8a'
9a'
7b'
3
7a'

FIG. 3

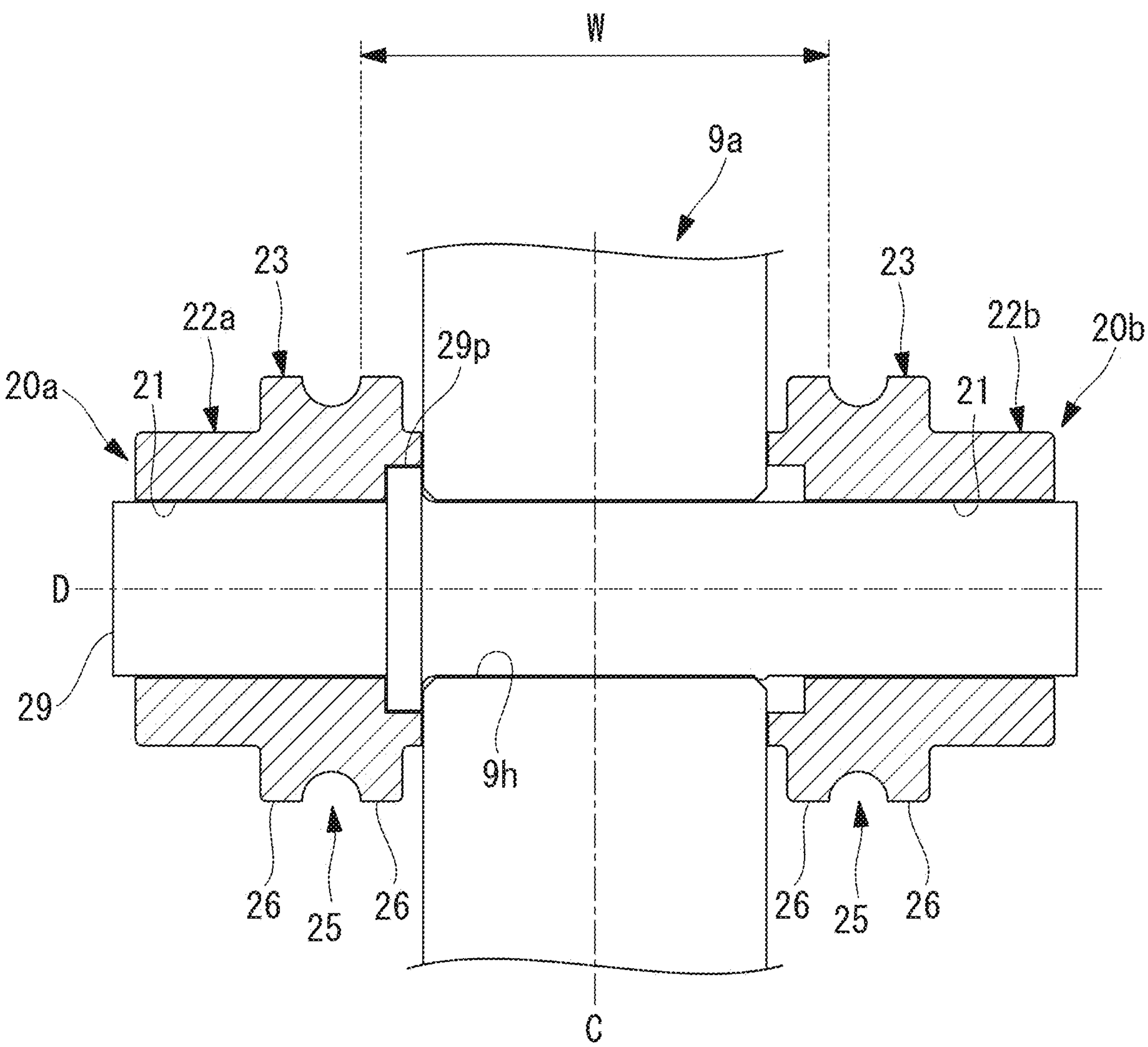


FIG. 4

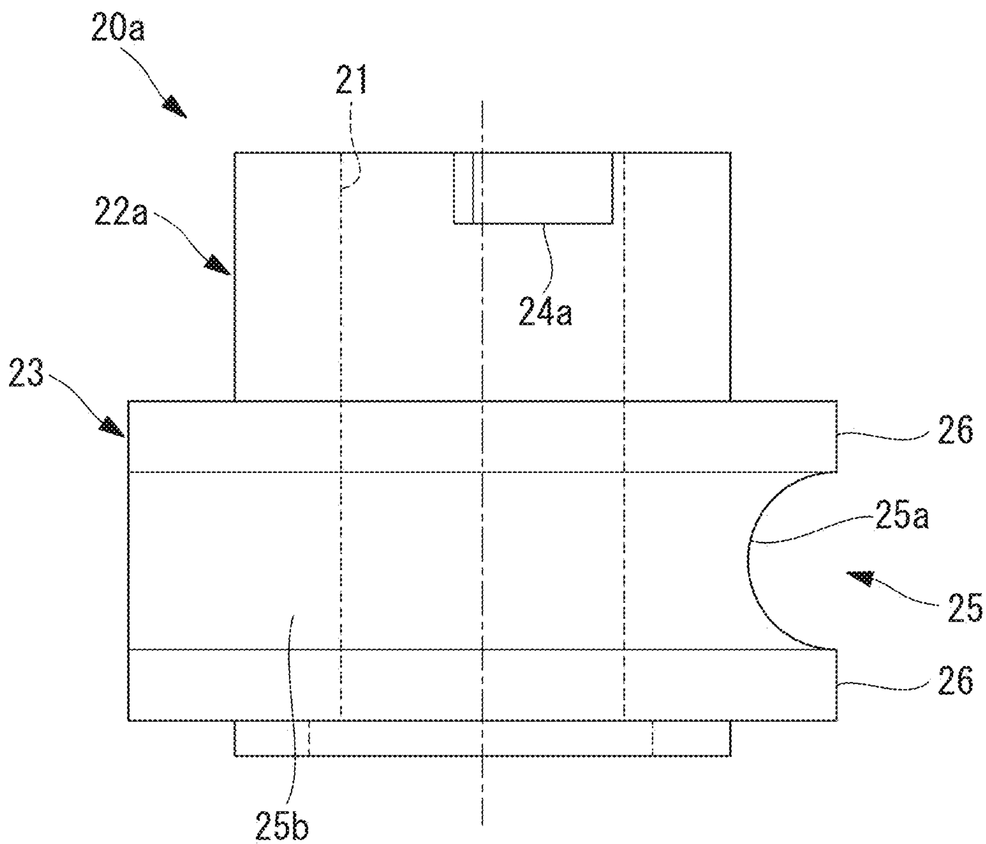


FIG. 5

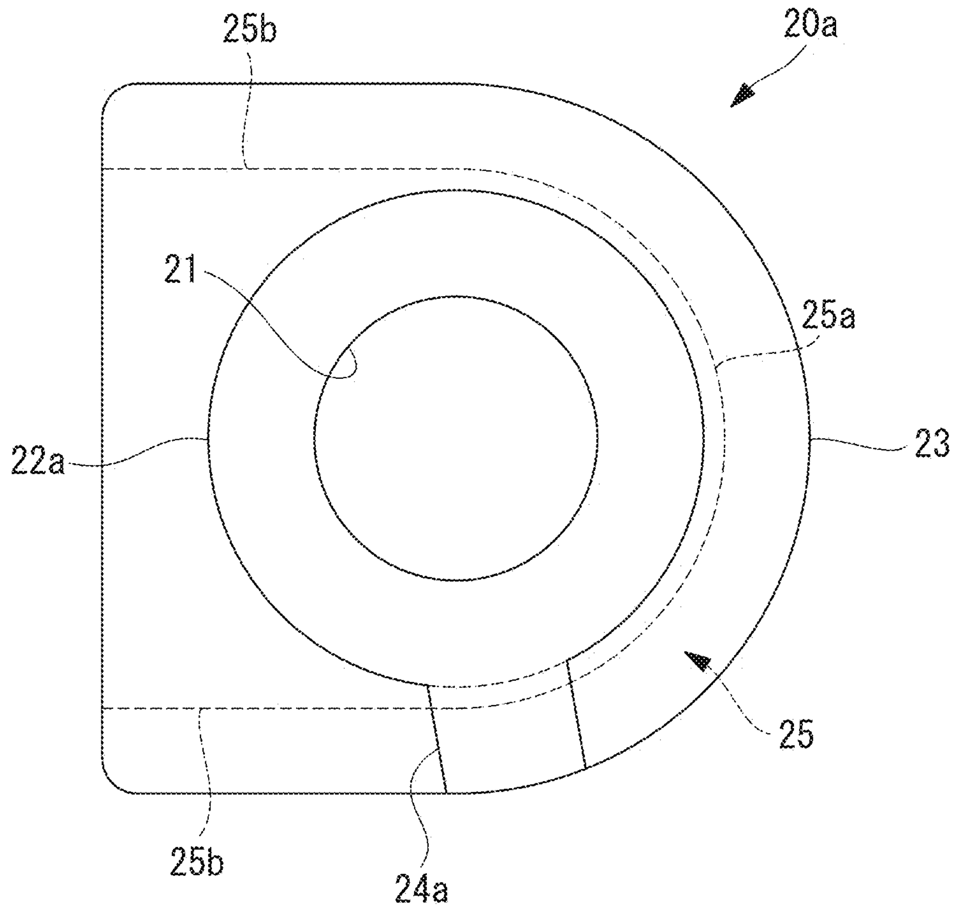


FIG. 6

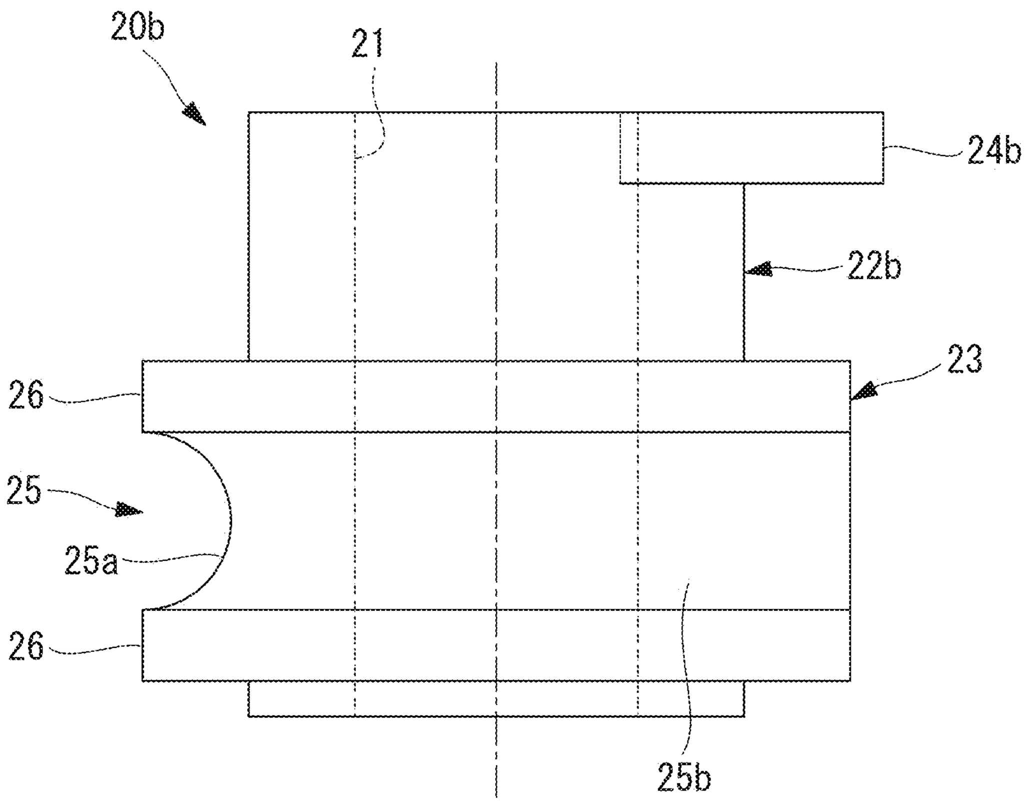


FIG. 7

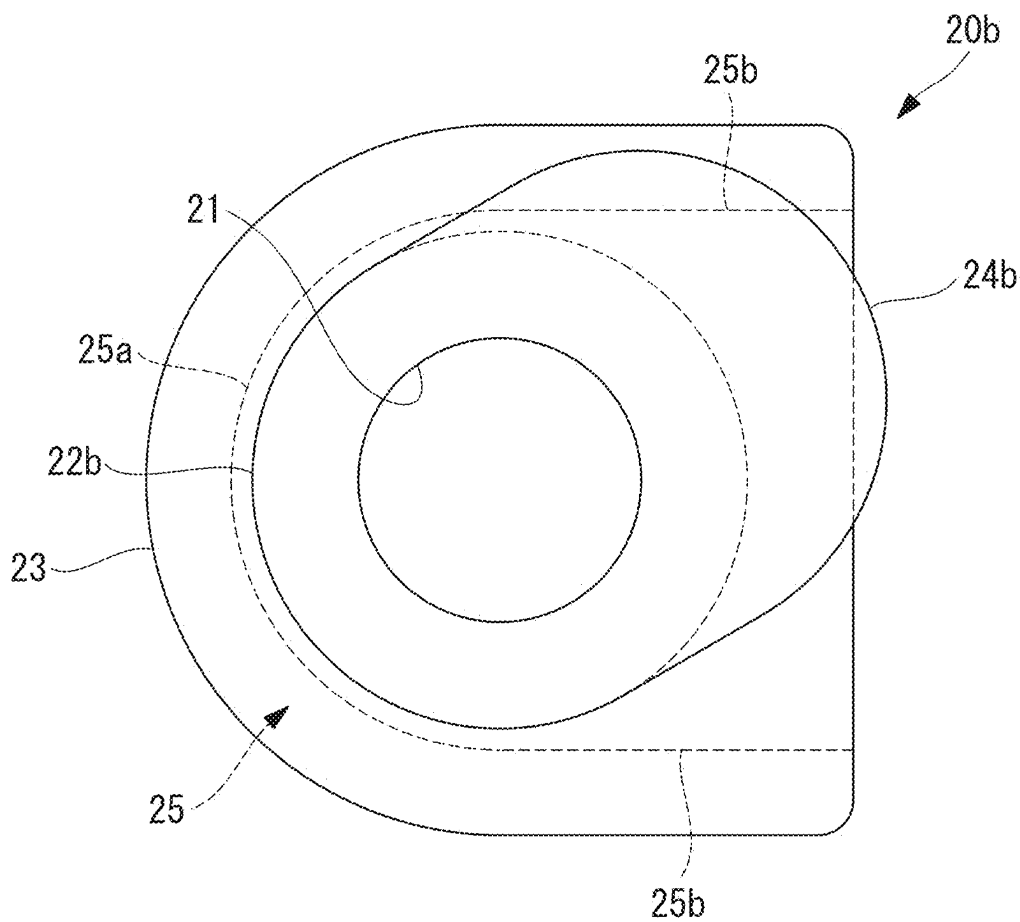


FIG. 8

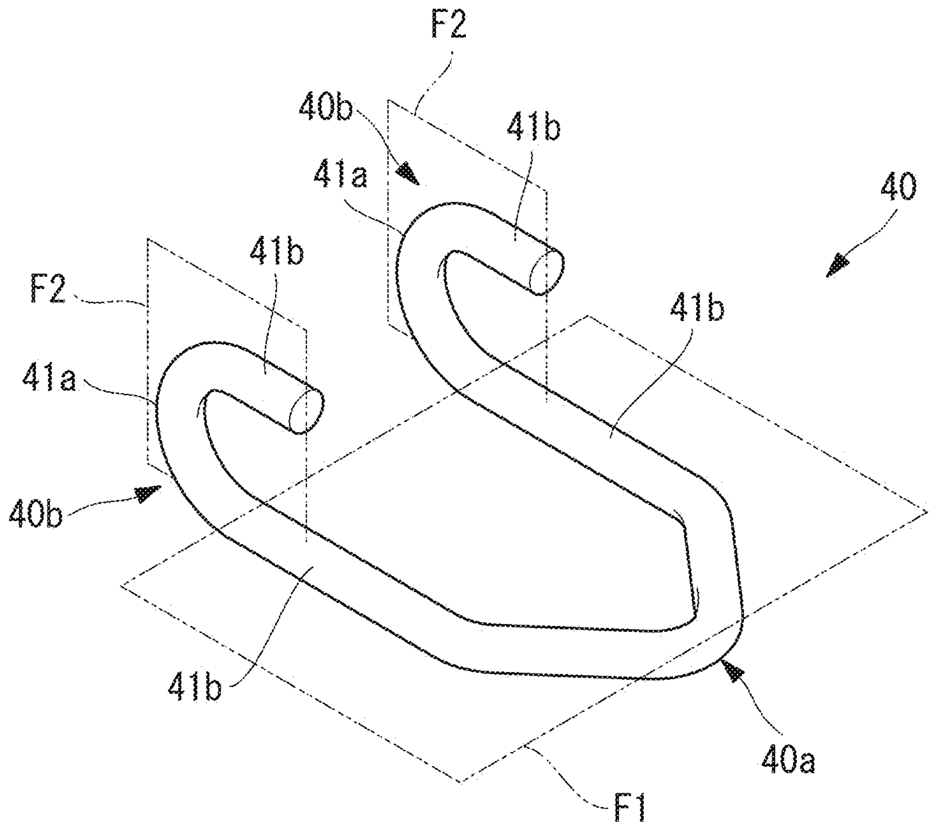


FIG. 9

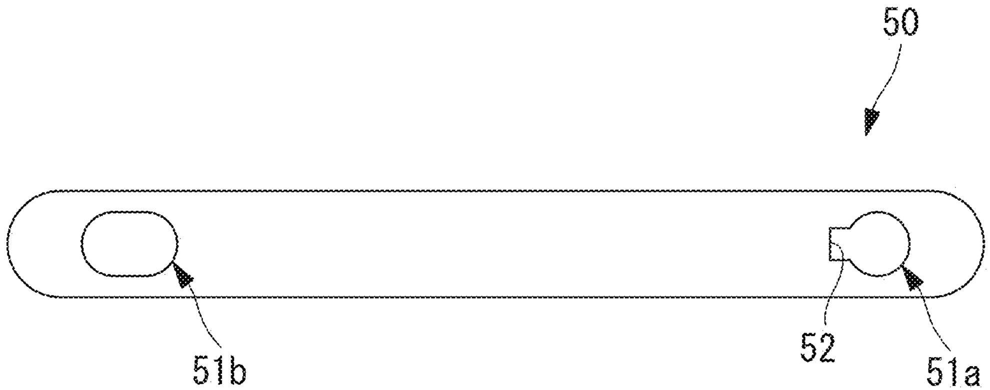
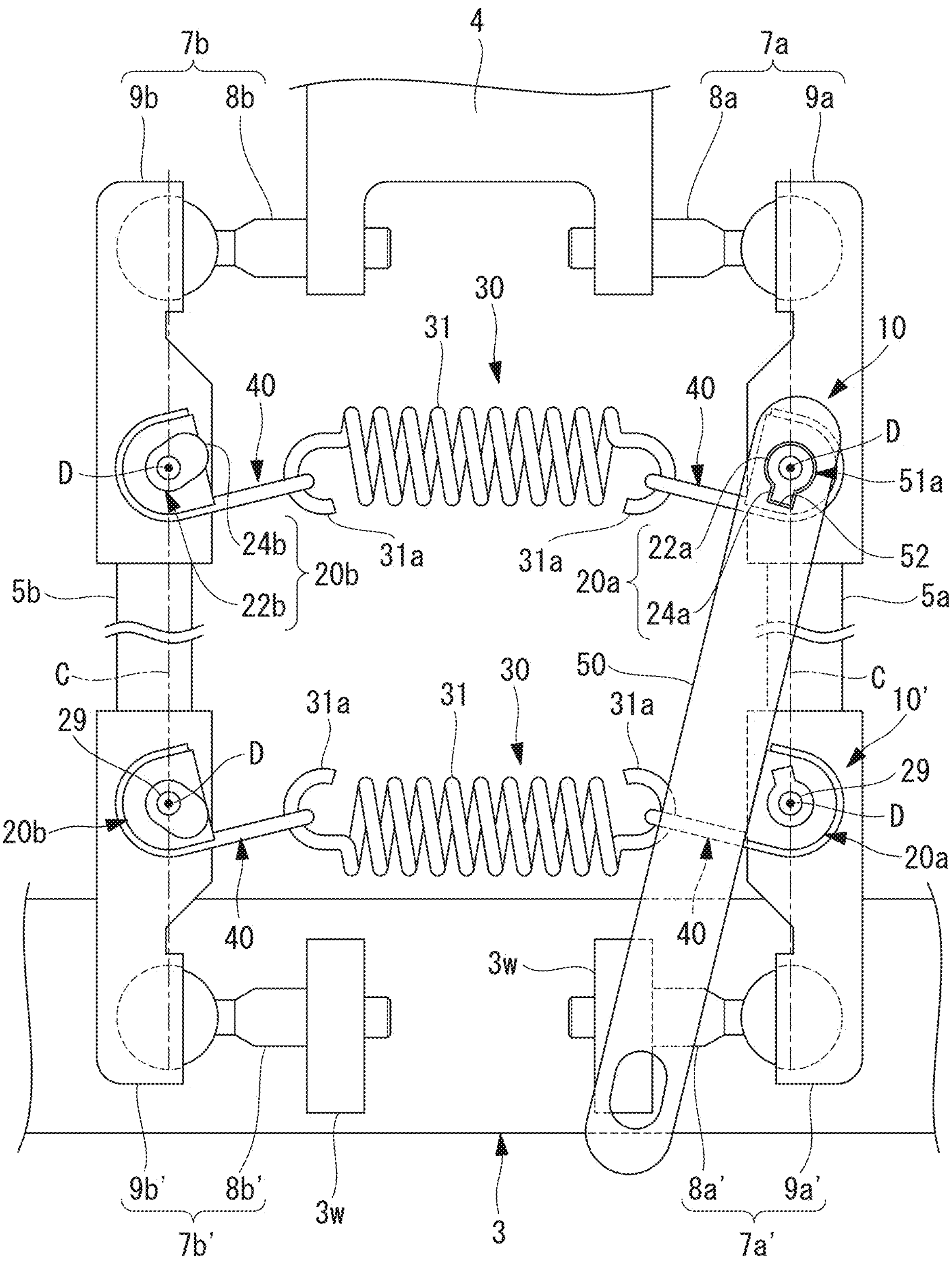


FIG. 10



4
7b
9b
8b
7a
8a
9a
30
31
10
40
40
D
D
51a
24b
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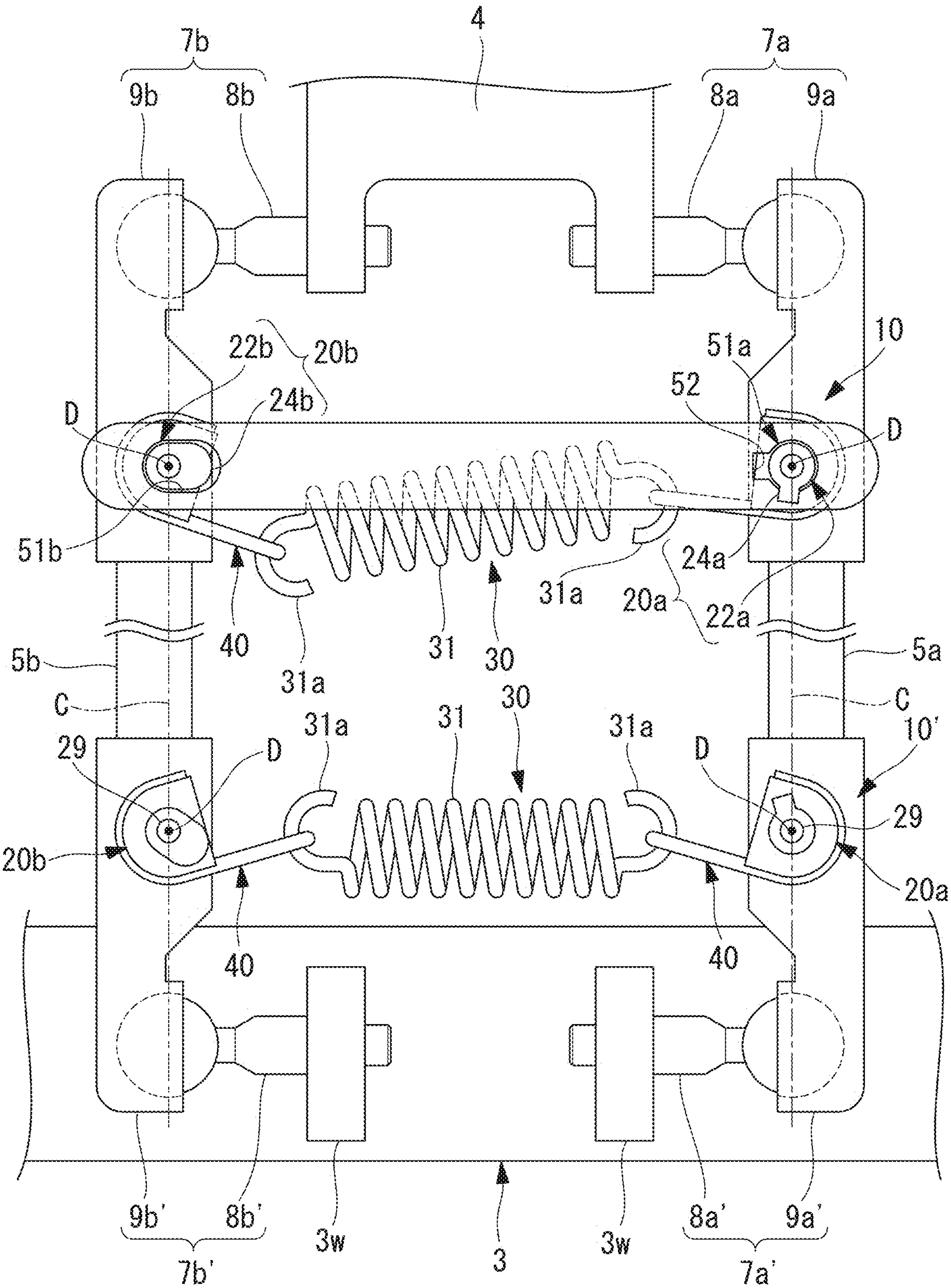
31a
22a
52
20b
20a
5b
22b
24a
5a
C
50
C
30
10'
31a
31
31a
29
29
D
D
20b
20a
40
40
3w
9b'
8b'
3w
8a'
9a'
7b'
3
7a'

FIG. 11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10
22b
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51a
52
24b
D
D
51b
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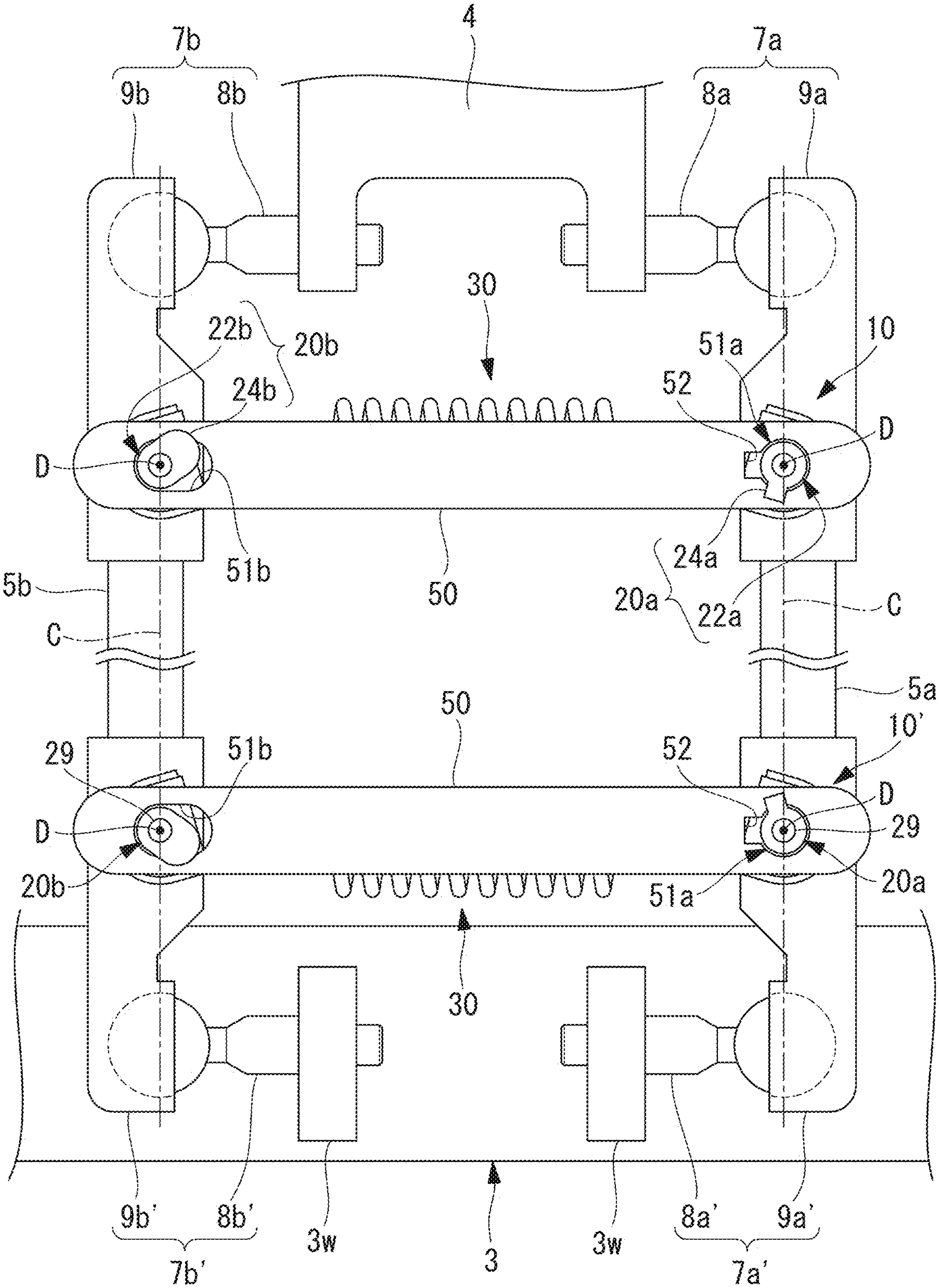
31
30
31a
24a
20a
22a
5b
5a
C
C
29
D
31a
31
30
31a
D
10'
29
20b
20a
40
40
9b'
8b'
3w
3w
8a'
9a'
7b'
3
7a'

FIG. 12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9a
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20b
24b
51a
10
52
D
D
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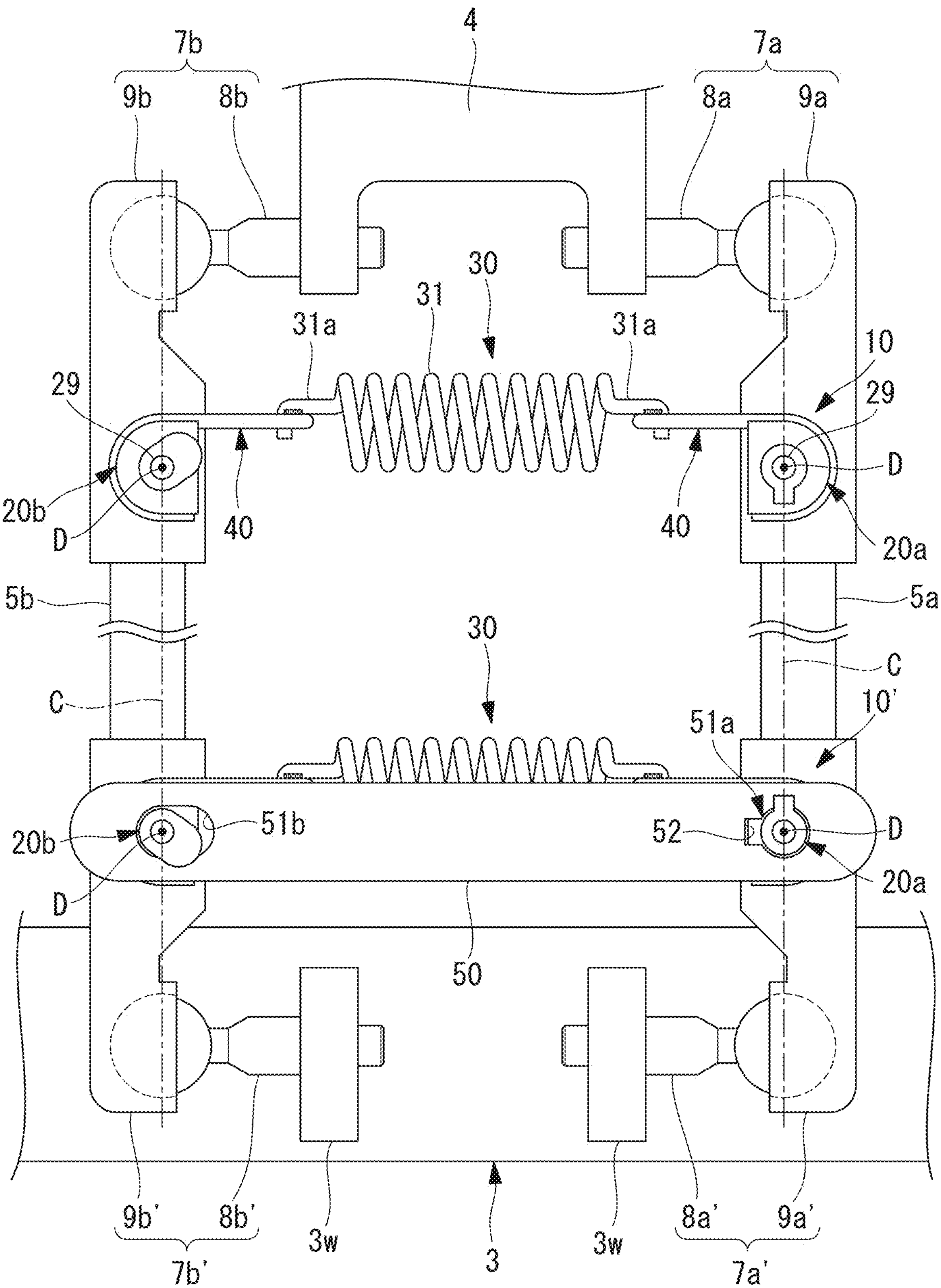
24a
20a
22a
5b
C
C
5a
50
10'
29
51b
52
D
D
29
20b
51a
20a
30
9b'
8b'
3w
3
3w
8a'
9a'
7b'
7a'

FIG. 13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7a
9b
8b
8a
9a
30
31
31a
31a
10
29
29
D
20b
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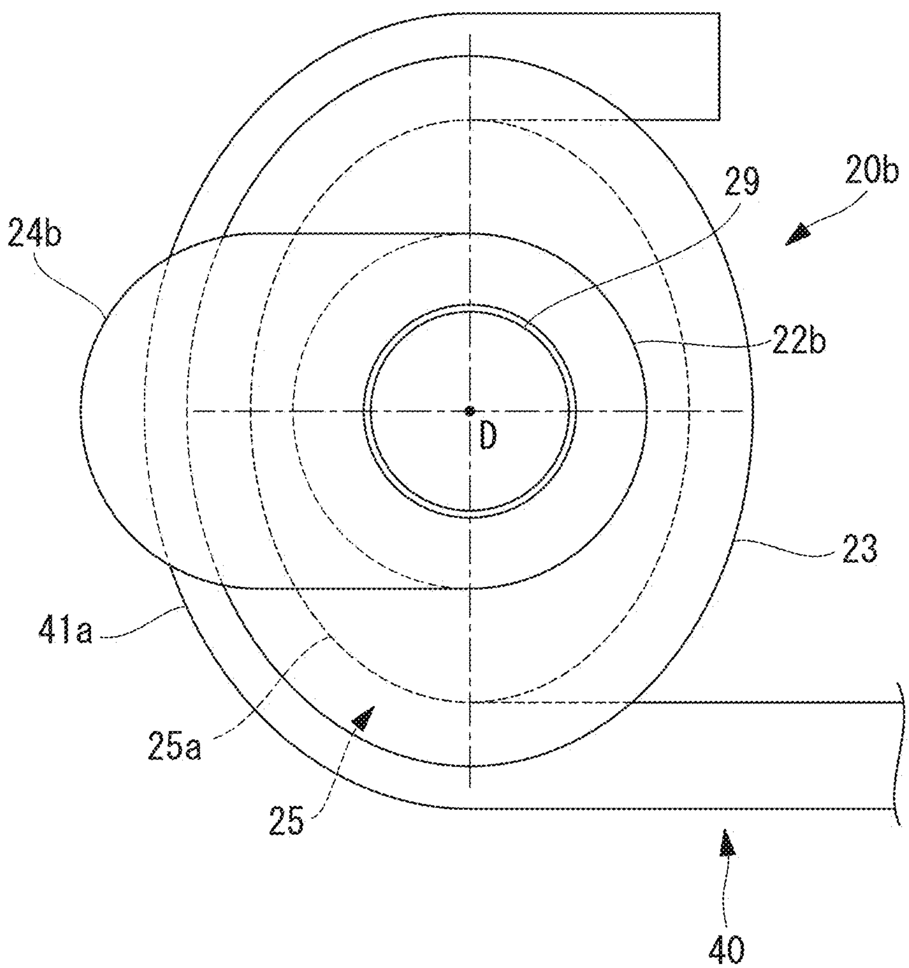
40
40
20a
5b
5a
30
C
C
51a
10'
20b
51b
52
D
20a
D
50
9b'
8b'
3w
3w
8a'
9a'
7b'
3
7a'

FIG. 14



CONNECTION STRUCTURE AND PARALLEL LINK ROBOT

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This is a National Stage Entry into the United States Patent and Trademark Office from International Patent Application No. PCT/JP2022/020116, filed on May 12, 2022, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present disclosure relates to a connection structure and a parallel link robot.

BACKGROUND ART

[0003] A spring device that connects two parallel links of a robot employing a parallel mechanism is known (for example, refer to Japanese Unexamined Patent Application Publication No. 2009-097571). This spring device restricts the distance between the two links from increasing to or beyond a prescribed distance while pulling the links closer together in the direction in which the links move closer together.

SUMMARY OF INVENTION

[0004] An aspect of the present disclosure provides a connection structure that interconnects two links. The two links have longitudinal axes and are spaced apart from each other in parallel and operated. The connection structure includes: bushings attached to the links so as to be rotatable about an axis perpendicular to a plane containing the two longitudinal axes; a biasing mechanism spanning between the bushings of the two links and applying an elastic restoring force to the two bushings in a direction in which the two bushings move closer together; and a connection member having two attachment holes through which the bushings can pass in a direction of the axis and regulating a distance between the bushings to be less than or equal to a prescribed distance. The bushings include claw portions projecting radially outward at part of, in a peripheral direction, an end portion of each of the bushings in the direction of the axis. The attachment holes are formed in a shape allowing the bushings to pass therethrough only at a prescribed attachment phase around the axis coinciding with the claw portions of the bushings. A phase of the bushings around the axis is regulated by the elastic restoring force to a phase that does not match the attachment phase.

BRIEF DESCRIPTION OF DRAWINGS

[0005] FIG. **1** is front view schematically illustrating a parallel link robot according to an embodiment of the present disclosure.

[0006] FIG. **2** is a front view schematically illustrating an attachment state of a biasing mechanism of a connection structure in a pair of links of the parallel link robot in FIG. **1**.

[0007] FIG. **3** is a sectional view illustrating the shapes of an attachment pin and a pair of bushings of one link of the parallel link robot in FIG. **1**.

[0008] FIG. **4** is a side view illustrating the shape of one bushing of the connection structure illustrated in FIG. **3**.

[0009] FIG. **5** is a front view illustrating the shape of the bushing illustrated in FIG. **4**.

[0010] FIG. **6** is a side view illustrating the shape of the other bushing of the connection structure illustrated in FIG. **3**.

[0011] FIG. **7** is a front view illustrating the shape of the bushing illustrated in FIG. **6**.

[0012] FIG. **8** is a perspective view illustrating the shape of a hook of the connection structure illustrated in FIG. **2**.

[0013] FIG. **9** is a front view illustrating the shape of a connection member of the connection structure illustrated in FIG. **2**.

[0014] FIG. **10** is a diagram for explaining the procedure of attaching the connection member in the connection structure illustrated in FIG. **2**.

[0015] FIG. **11** is a diagram for explaining the procedure of attaching the connection member in the connection structure illustrated in FIG. **2**.

[0016] FIG. **12** is a front view schematically illustrating an attachment state of the connection member in the connection structure illustrated in FIG. **2**.

[0017] FIG. **13** is a front view schematically illustrating the shape of a biasing mechanism in a first modification of the connection structure illustrated in FIG. **2**.

[0018] FIG. **14** is a schematic diagram illustrating the shape of one bushing in a second modification of the connection structure illustrated in FIG. **2**.

DESCRIPTION OF EMBODIMENT(S) OF THE INVENTION

[0019] A connection structure **10** and a parallel link robot **1** according to an embodiment of the present disclosure will be described below while referring to the drawings.

[0020] The parallel link robot **1** according to this embodiment includes, for example, a base **2** that is suspended from and is fixed to a ceiling or the like, as illustrated in FIG. **1**. The parallel link robot **1** further includes a movable part **3** disposed below and spaced apart from the base **2**. Furthermore, the parallel link robot **1** includes three arms **4**. The arms **4** are supported so as to be rotatable around three respective horizontal axes A with respect to the base **2**. The parallel link robot **1** further includes a pair of mutually parallel links **5***a*, **5***b* for each arm **4**. Each pair of parallel links **5***a*, **5***b* connects the corresponding arm **4** to the movable part **3**. The parallel link robot **1** further includes connection structures **10**, **10'** spanning between each pair of links **5***a*, **5***b*.

[0021] The base **2** includes three servomotors **6** for respectively driving the three arms **4**. The servomotors **6** are disposed at equal intervals along a peripheral direction around an axis B that passes through the center of the base **2** and extends in the vertical direction.

[0022] The servomotors **6** each include a rotary drive shaft (not illustrated) that is made to rotate around the horizontal axis A.

[0023] Each arm **4** is connected at its base end to the rotary drive shaft of the corresponding servomotor **6** and can rotate around the axis A relative to the base **2**. As illustrated in FIG. **2**, the parallel link robot **1** further includes pairs of ball joints **7***a*, **7***b*. Each pair of ball joints **7***a*, **7***b* connects the leading end of the corresponding arm **4** to the corresponding links **5***a*, **5***b*. The ball joints **7***a*, **7***b* include ball studs **8***a*, **8***b* and sockets **9***a*, **9***b*.

[0024] The ball studs **8***a*, **8***b* are externally attached to both sides of the leading end of each arm **4** in the direction of the axis A (see FIG. **1**) with the leading end of the arm **4** interposed therebetween.

[0025] Each pair of links **5***a*, **5***b* includes the sockets **9***a*, **9***b* at one end thereof in the direction of the respective longitudinal axis C. The sockets **9***a*, **9***b* are respectively fitted to the ball studs **8***a*, **8***b* at the leading end of corresponding arm **4** as through being sandwiched from both sides in the direction of the axis A.

[0026] Each socket **9***a*, **9***b* includes an attachment pin **29** protruding in a direction perpendicular to the longitudinal axis C, as illustrated in FIGS. **2** and **3**. Each attachment pin **29** is press-fitted into a pin hole **9***h* that extends through the socket **9***a*, **9***b* in a direction perpendicular to the longitudinal axis C, and protrudes on both sides of the longitudinal axis C. The attachment pin **29** is press-fitted into the pin hole **9***h* from one side until a flange-shaped positioning portion **29***p* provided at a midway position along the length direction is abutted. In this way, the amounts by which the attachment pin **29** protrudes on both sides of the outer peripheral surface of the socket **9***a*, **9***b* are adjusted to be the same length as each other. In the assembled state, the attachment pin **29** extends along an axis D perpendicular to the plane containing the longitudinal axes C of the two links **5***a*, **5***b*.

[0027] Each link **5***a*, **5***b* includes a socket **9***a*', **9***b*', which is similar to that at the one end, at the other end thereof in the direction of the longitudinal axis C. The sockets **9***a*', **9***b*' together with ball studs **8***a*', **8***b*' provided on the movable part **3** constitute ball joints **7***a*', **7***b*'. Similarly to the sockets **9***a*, **9***b*, the sockets **9***a*', **9***b*' each include the pin hole **9***h* and the attachment pin **29** press-fitted into the pin hole **9***h*.

[0028] The movable part **3** includes a pair of parallel flat-plate-like attachment portions **3***w* protruding radially outward from the outer peripheral surface of the movable part **3** at positions corresponding to each pair of links **5***a*, **5***b*. The ball studs **8***a*', **8***b*' are attached to the outside of each pair of attachment portions **3***w*. The sockets **9***a*', **9***b*' at the other end of the links **5***a*, **5***b* are respectively fitted to the ball studs **8***a*', **8***b*' as though being sandwiched from both sides in the direction of the axis A.

[0029] Next, the connection structures **10**, **10**' according to this embodiment will be described.

[0030] In this embodiment, the three pairs of links **5***a*, **5***b* and the three upper and lower connection structures **10**, **10**' each have substantially the same configuration. Therefore, the configurations of one pair of links **5***a*, **5***b* and one connection structure **10** will be described below as an example.

[0031] The connection structure **10** includes bushings **20***a*, **20***b* attached to the sockets **9***a*, **9***b*, as illustrated in FIGS. **2** and **3**, for example. The connection structure **10** includes a biasing mechanism **30** and connection members **50**, which span between the sockets **9***a*, **9***b* and connect the sockets **9***a*, **9***b* to each other, as illustrated in FIG. **2**.

[0032] The bushings **20***a*, **20***b* are, for example, manufactured by injection molding a resin material such as silicone resin. The bushings **20***a*, **20***b* each have a cylindrical form including a hollow hole **21**, as illustrated in FIGS. **4** to **7**.

[0033] The hollow hole **21** has a circular cross section that allows the attachment pin **29** to be rotatably fitted thereinto. The hollow hole **21** in the bushing **20***a* is provided with a relief portion corresponding to the positioning portion **29***p* provided on the attachment pin **29**. Thus, the combination of bushings **20***a*, **20***b* to be attached to the two ends of the attachment pin **29** is uniquely determined.

[0034] The bushings **20***a*, **20***b* respectively include first cylindrical portions **22***a*, **22***b* on one side in the axial direction and second cylindrical portions **23** on the other side in the axial direction. The first cylindrical portions **22***a*, **22***b* respectively include, at the leading ends of the cylindrical parts in the axial direction, claws (claw portions) **24***a*, **24***b* formed so as to protrude radially outward along part of the peripheral direction.

[0035] The claws **24***a*, **24***b* will be described later.

[0036] The second cylindrical portions **23** have larger outer shapes than an outer diameter dimensions of the first cylindrical portions **22***a*, **22***b*. Each second cylindrical portion **23** includes a groove **25** that is radially recessed at a position midway along the respective axial direction. Thus, the second cylindrical portion **23** includes flanges **26** that extend radially outward from the groove **25** on both sides of the groove **25** in the respective axial direction.

[0037] The groove **25** includes a semi-circular arc-shaped curved groove **25***a* and a pair of straight grooves **25***b*, which are parallel to each other and are continuous with the two ends of the curved groove **25***a*. As a result, the groove **25** and the flanges **26** have a non-circular cross-sectional shape.

[0038] The biasing mechanism **30** includes a coil spring (elastic body) **31** and two hooks **40** disposed at the two ends of the coil spring **31**, as illustrated in FIG. **2**. The hooks **40** are connected to the two ends of the coil spring **31** by hooking onto hook portions **31***a* at the two ends of the coil spring **31**.

[0039] The hook portions **31***a* each have a shape obtained by bending the wire constituting the coil spring **31** into a hook shape. The coil spring **31** has a spring constant of sufficient size to apply a preload of an appropriate magnitude to the ball joints **7***a*, **7***b*.

[0040] Each hook **40** is a member obtained by bending and shaping a round metal bar, as illustrated in FIG. **8**. The round metal bar has an outer diameter dimension equivalent to the groove width of the grooves **25** of the bushings **20***a*, **20***b*.

[0041] Each hook **40** includes a first connection portion **40***a* formed by bending the hook **40** at the center in the length direction along any first plane F**1**. The hook **40** further includes a pair of second connection portions **40***b* formed by bending both ends of the hook **40** in the length direction along a second plane F**2** perpendicular to the first plane F**1**. Each second connection portion **40***b* includes a pair of straight portions **41***b* located at the two ends of a semicircular arc-shaped curved portion **41***a* and extending parallel to each other.

[0042] The spacing between the pair of second connection portions **40***b* is equivalent to a distance W between the grooves **25** of the two bushings **20***a*, **20***b* attached to the socket **9***a* (see FIG. **3**).

[0043] Each of the second connection portions **40***b* has a form that matches the shape of the corresponding groove **25** around the axis D.

[0044] The pair of second connection portions **40***b* are respectively hooked onto the grooves **25** of the two bushings **20***a*, **20***b* attached to the socket **9***a*. The curved portion **41***a* and the two straight portions **41***b* of each second connection portion **40***b* then closely contact the bottom of the corresponding groove **25**. In other words, the curved portions **41***a* of each hook **40** closely contact the corresponding curved grooves **25***a* and the straight portions **41***b* of the hook **40**

closely contact the corresponding straight grooves **25***b*. In this way, the two bushings **20***a*, **20***b* and the corresponding hook **40** are connected in an integrated manner so as to not rotate relative to each other around the axis D of the attachment pin **29**.

[0045] The hook portions **31***a* of the coil spring **31** are hooked onto the first connection portions **40***a* of the hooks **40** with the second connection portions **40***b* hooked onto the bushings **20***a*, **20***b*. When the second connection portions **40***b* are hooked onto the grooves of the bushings **20***a*, **20***b*, the first connection portions **40***a* of the hooks **40** extend parallel to the axes D of the attachment pins **29**. Therefore, the hook portions **31***a* at both ends of the coil spring **31** are hooked onto the two hooks **40** so as to be rotatable about axes parallel to the axes D.

[0046] As illustrated in FIG. **9**, each connection member **50** is a long strip-shaped member formed by punching a flat metal plate. The connection members **50** are disposed so that one connection member **50** is disposed on each side of the sockets **9***a*, **9***b* in the direction of the axis D. Each connection member **50** includes attachment holes **51***a*, **51***b* that penetrate therethrough in the plate thickness direction in the vicinity of both ends of the connection member **50** in the length direction.

[0047] The attachment hole **51***a* has an opening shape that allows the first cylindrical portion **22***a*, including the claw **24***a* of the bushing **20***a*, to pass therethrough in the axial direction of the first cylindrical portion **22***a*.

[0048] The claw **24***a* has a width dimension that is smaller than the diameter of the cylindrical part of the first cylindrical portion **22***a* and a length dimension that projects radially up to a position equivalent to the outer shape of the flange **26**. The claw **24***a* is shaped like a flat plate having a prescribed thickness. The claw **24***a* extends from the outer surface of the cylindrical part of the first cylindrical portion **22***a* in a direction different from the length direction of the straight groove **25***b*, for example, in a direction at an angle of 120°.

[0049] The attachment hole **51***a* includes a cutout **52**, along part of the circumferential direction of the circular opening shape having an inner diameter dimension slightly larger than the outer diameter dimension of the cylindrical part of the first cylindrical portion **22***a*. The size and shape of the cutout **52** are similar to the shape of and slightly larger than the claw **24***a* so as to allow the claw **24***a* to pass therethrough.

[0050] The attachment hole **51***b* has a long hole-shaped opening shape that allows the first cylindrical portion **22***b*, including the claw **24***b* of the bushing **20***b*, to pass therethrough in the axial direction of the first cylindrical portion **22***b*. The claw **24***b* has a width dimension that is equivalent to the diameter of the cylindrical part of the first cylindrical portion **22***a* and a length dimension that projects radially further than the outer diameter of the flange **26**. The claw **24***b* is shaped like a flat plate having a prescribed thickness. The claw **24***b* extends from the outer surface of the cylindrical part of the first cylindrical portion **22***b* in a direction different from the length direction of the straight groove **25***b*, for example, in a direction at an angle of 30°.

[0051] The attachment hole **51***b* has a width dimension slightly larger than the width dimension of the claw **24***b* and a length dimension slightly larger than the dimension of the first cylindrical portion **22***b* in the length direction of the claw **24***b*. The attachment hole **51***b* extends in the longitudinal axial direction of the connection member **50** and allows the cylindrical part of the first cylindrical portion **22***b* to pass therethrough so as to be movable in the longitudinal axial direction of the connection member **50**.

[0052] The connection member **50** allows the first cylindrical portions **22***a*, **22***b* disposed on the same side of the sockets **9***a*, **9***b* in the direction of the axis D to pass through the attachment holes **51***a*, **51***b*, respectively.

[0053] In this case, the phases around the D axis of the length direction of the claws **24***a*, **24***b*, which pass through the attachment holes **51***a*, **51***b*, and the length direction of the cutout **52** are different from each other.

[0054] Operation of the thus-configured parallel link robot **1** and connection structure **10** will be described hereafter. In the following description, one of the two connection members **50** will be described as an example, and the description of the other connection member **50** will be omitted.

[0055] The parallel link robot **1** according to this embodiment is assembled as follows.

[0056] In the assembly, a base unit, the movable part **3**, three pairs of links **5***a*, **5***b*, and three of each of the connection structures **10**, **10'** are prepared. The base unit is obtained by assembling three arms **4** and the ball studs **8***a*, **8***b* from the base **2**. Each link **5***a*, **5***b* has the socket **9***a*, **9***b* and the attachment pin **29** attached at one end and the socket **9***a'*, **9***b'* and the attachment pin **29** attached at the other end.

[0057] In order to assemble the parallel link robot **1**, first, the two hooks **40** are respectively connected to the coil springs **31** of the connection structures **10**, **10'**. Specifically, the hook portions **31***a* on both sides of each coil spring **31** are hooked onto the first connection portions **40***a* of the two hooks **40**, respectively.

[0058] Next, a pair of bushings **20***a*, **20***b* are attached to the attachment pins **29** of the sockets **9***a*, **9***b* of the links **5***a*, **5***b*. The bushings **20***a*, **20***b* are brought closer together from both sides of the sockets **9***a*, **9***b*, and the attachment pins **29** are fitted into the hollow holes **21**. Thus, the pair of bushings **20***a*, **20***b* are attached so as to be rotatable around the axis D of the same attachment pin **29**.

[0059] Similarly, a pair of bushings **20***a*, **20***b* are attached to the attachment pin **29** of each socket **9***a'*, **9***b'* of each link **5***a*, **5***b*. The bushings **20***a*, **20***b* are respectively attached to both ends of the attachment pin **29** of each socket **9***a'*, **9***b'* so as to be rotatable around the axis D of the attachment pin **29**.

[0060] Next, the pair of second connection portions **40***b* of the hook **40** on one side of the connection structure **10** are hooked onto the grooves **25** of the bushings **20***a*, **20***b* attached to the socket **9***a*. The pair of second connection portions **40***b* of the hook **40** on the other side are hooked onto the grooves **25** of the bushings **20***a*, **20***b* attached to the socket **9***b*.

[0061] Similarly, the hook **40** on one side of the connection structure **10'** is hooked onto the grooves **25** of the bushings **20***a*, **20***b* attached to the socket **9***a'*. The hook **40** on the other side is hooked onto the grooves **25** of the bushings **20***a*, **20***b* attached to the socket **9***b'*.

[0062] Next, the sockets **9***a*, **9***a'* of one of the pair of links **5***a*, **5***b*, for example, the link **5***a* are respectively brought closer to the ball studs **8***a*, **8***a'*. The sockets **9***a*, **9***a'* are then fitted into the ball studs **8***a*, **8***a'* from the outside in the direction of the axis A. Then, while holding the link **5***b* on the other side, the two coil springs **31** are pulled in the direction of extension in order to increase the distance between the links **5***a*, **5***b*. Then, the sockets **9***b*, **9***b'* of the link

5*b* are simultaneously fitted into the ball studs **8***b*, **8***b*', respectively, from the outside in the direction of the axis A. As a result, the ball studs **8***a*, **8***b* are sandwiched in the direction of the axis A between the pair of sockets **9***a*, **9***b*. Similarly, the ball studs **8***a*', **8***b*' are sandwiched in the direction of the axis A between the pair of sockets **9***a*', **9***b*'.

[0063] In this way, the pair of links **5***a*, **5***b* are connected to each other by the two biasing mechanisms **30** and attached to the arm **4** and the movable part **3**.

[0064] In this state, the two second connection portions **40***b* of one hook **40** are hooked onto the grooves **25** of the two bushings **20***a*, **20***b* with the socket **9***a*, **9***b* interposed therebetween. Thus, the hook **40** is integrally connected to the pair of bushings **20***a*, **20***b*.

[0065] The flanges **26** next to both sides of each groove **25** are positioned so that the corresponding second connection portion **40***b* of the hook **40** is interposed therebetween in the axial direction of the second cylindrical portions **23**. In this way, the bushings **20***a*, **20***b* restrict the movement of the hook **40** along the axial direction of the attachment pin **29**. The hook **40** also restricts the bushings **20***a*, **20***b* from falling off the attachment pin **29**. In other words, the bushings **20***a*, **20***b* and the hook **40** are rotatable in an integrated manner around the axis D of the attachment pin **29** and do not move relative to each other in any other direction. This similarly applies to the two bushings **20***a*, **20***b* with the socket **9***a*', **9***b*' interposed therebetween.

[0066] Next, the connection member **50** is attached between the links **5***a*, **5***b*.

[0067] As illustrated in FIG. **10**, the first cylindrical portion **22***a* of the bushing **20***a* attached to one link **5***a* is passed through the attachment hole **51***a* of the connection member **50**. By aligning the length direction of the cutout **52** of the attachment hole **51***a* with the phase of the claw **24***a* of the first cylindrical portion **22***a*, the first cylindrical portion **22***a* is easily passed through the attachment hole **51***a*. Once the claw **24***a* has passed through the attachment hole **51***a*, the connection member **50** is rotated around the axis D of the bushing **20***a*. In this way, the phases of the attachment hole **51***a* and the claw **24***a* are shifted from each other, and therefore the connection member **50** is attached to the first cylindrical portion **22***a* of the bushing **20***a* so as to not come off.

[0068] Next, the connection member **50** is rotated around the axis D of the bushing **20***a* up to the position where the attachment hole **51***b* is aligned with the first cylindrical portion **22***b* of the bushing **20***b*. In this state, the phase of the claw **24***b* of the bushing **20***b* is not aligned with the length direction of the attachment hole **51***b* of the connection member **50**.

[0069] Then, as illustrated in FIG. **11**, force is applied to the connection part between the hook portion **31***a* of the coil spring **31** and the first connection portion **40***a* of the hook **40**. This causes the coil spring **31** and the hook **40** to rotate relative to each other, the coil spring **31** stretches slightly, and the hook **40** and the bushing **20***b* rotate about the axis D.

[0070] At the time point when the bushing **20***b* has been rotated until the phase of the claw **24***b* is aligned with the length direction of the attachment hole **51***b*, the first cylindrical portion **22***b* is passed through the attachment hole **51***b*. At the time point when the claw **24***b* has passed through the attachment hole **51***b*, the force applied to the coil spring **31** is released. Thus, as illustrated in FIG. **12**, the phases of the claw **24***b* and the attachment hole **51***b* are shifted from each other, and the connection member **50** is attached to the first cylindrical portion **22***b* so as to not come off.

[0071] Similarly, the connection member **50** of the connection structure **10'** is suspended between the bushings **20***a*, **20***b* attached to the attachment pins **29** of the sockets **9***a*', **9***b*'. In the example illustrated in FIG. **10**, the claw **24***b* of the bushing **20***b* of the connection structure **10'** extends in a different direction from the claw **24***b* of the connection structure **10**. This ensures that the coil spring **31** does not interfere with surrounding members when inserting the bushing **20***b* through the attachment hole **51***b* of the connection member **50** of the connection structure **10'**.

[0072] The remaining two pairs of links **5***a*, **5***b* can be assembled in substantially the same manner, and in this way, the assembly of the parallel link robot **1** is completed.

[0073] Thus, according to this embodiment, a force is applied in the direction in which the spacing between the pair of links **5***a*, **5***b* becomes smaller and it is possible to restrict the links **5***a*, **5***b* from becoming separated beyond a prescribed amount. There is advantage in that the connection member **50**, which regulates widening of the spacing, can be assembled or removed without the use of tools or jigs.

[0074] In addition, the bushings **20***a*, **20***b*, which support the connection member **50** with respect to the pair of links **5***a*, **5***b*, are held in prescribed postures by the biasing mechanism **30**. In other words, the claws **24***a*, **24***b* and the attachment holes **51***a*, **51***b* of the connection member **50** are positioned with phases so as to prevent the connection member **50** from falling off. This prevents the connection member **50** from unintentionally falling off such as during the operation of the links **5***a*, **5***b*.

[0075] One attachment hole **51***a* of the connection member **50** has a circular cross section, and the first cylindrical portion **22***a* that passes through the attachment hole **51***a* does not undergo any relative movement in the radial direction inside the attachment hole **51***a*. Therefore, the parallel link robot **1** can operate without causing rattling between the connection member **50** and the bushing **20***a*.

[0076] Since one attachment hole **51***b* of the connection member **50** is a long hole, the first cylindrical portion **22***b* is allowed to undergo some displacement while remaining passing through the attachment hole **51***b*. This has the advantage that even if the distance between the links **5***a*, **5***b* decreases due to aging of the ball joints **7***a*, **7***b*, the decrease can be tolerated.

[0077] In this embodiment, the two hooks **40** are rotatably connected to the hook portions **31***a* at both ends of the coil spring **31** around an axis parallel to the axis D. Alternatively, at least one of the hooks **40** may be integrally fixed to the corresponding hook portion **31***a*.

[0078] For example, the hook **40** on the bushing **20***a* side may be secured to the hook portion **31***a* using welding or the like. If the hook **40** on the bushing **20***b* side and the coil spring **31** can rotate relative to each other, the phase of the claw **24***b* can be easily made to match the phase of the attachment hole **51***b*.

[0079] The two hooks **40** can be welded to the hook portions **31***a* on the two sides of the coil spring **31**, as illustrated in FIG. **13**. In this case, the assembly may be carried out in a different order from that in the above-described assembly method. For example, first, the connection member **50** is connected to the bushings **20***a*, **20***b*. Next, the hooks **40** welded on both sides can be hooked onto the bushings **20***a*, **20***b* while stretching the coil spring **31**.

[0080] In other words, the rotation of the bushings **20***a,* **20***b* around the axis D is not restricted when the biasing mechanism **30** is not attached. Therefore, the connection member **50** is attached by simultaneously aligning the phases of the claws **24***a,* **24***b* with the phases of the two attachment holes **51***a,* **51***b* of the connection member **50**. Then, the bushings **20***a,* **20***b* are rotated about the axis D so as to change the phases. After that, the biasing mechanism **30** is attached in order to regulate the rotation of the bushings **20***a,* **20***b.*

[0081] This allows the coil spring **31** and the two hooks **40** to be integrally handled. Assembly work can be made easier when the space in which the connection structure **10** is to be attached is small.

[0082] In this case, the phases of the claws **24***a,* **24***b* with respect to the straight grooves **25***b* can be freely set. Therefore, the claw **24***b* can also be disposed in a phase that is significantly displaced from the length direction of the attachment hole **51***b* to more reliably prevent the connection member **50** from falling off.

[0083] In particular, if the phases of the claws **24***a,* **24***b* with respect to the straight grooves **25***b* are the same, the same bushing as the bushing **20***a* can be used as the bushing **20***b.* As a result, the number of components can be reduced and manufacturing costs can be reduced by standardizing the components. In addition, during assembly work, there is no need to pay attention to the type of bushings and work efficiency can be improved.

[0084] In this case, by maintaining the shape of the attachment hole **51***b* as a long hole, changes in the distance between the links **5***a,* **5***b* due to aging etc. can be accommodated.

[0085] In this embodiment, the contact between the straight portions **41***b* of the hooks **40** and the straight grooves **25***b* of the bushings **20***a,* **20***b* regulates the relative rotation therebetween. Alternatively, the bushings **20***a,* **20***b* do not have to include the straight grooves **25***b.*

[0086] For example, as illustrated in FIG. **14**, the groove **25** of the bushing **20***b* has a non-circular cross-sectional shape, such as an oval shape. The curved portions **41***a* of the hooks **40** are curved into a shape that matches the grooves **25** of the bushings **20***a,* **20***b.*

[0087] As a result, when the biasing mechanism **30** is connected, the rotation of the bushing **20***b* around the axis D is regulated by the elastic restoring force of the coil spring **31**. As a result, the phases around the axis D of the claw **24***b* of the bushing **20***b* and the attachment hole **51***b* of the connection member **50** can be maintained shifted from each other similarly to as described above.

[0088] Substantially the same configuration can also be used for bushing **20***a* and the hook **40** connected to bushing **20***a* to achieve the same effect.

[0089] The cross-sectional shape of the grooves **25** in the bushings **20***a,* **20***b* may be noncircular shapes other than an oval shape. Any cross-sectional shape may be adopted as long as the shape regulates rotation around the axis D by contacting the curved portions **41***a* of the hook **40**.

[0090] For example, the cross-sectional shape of the grooves **25** of the bushings **20***a,* **20***b* may be a circular shape decentered with respect to the axis D. This also allows the rotation of bushings **20***a,* **20***b* around the axis D to be regulated by contacting the curved portions **41***a* of the hook **40**.

[0091] According to this embodiment, one attachment pin **29** was press-fitted into each of the sockets **9***a,* **9***b,* with both ends of the attachment pin **29** protruding in the direction of the axis D. Alternatively, two attachment pins **29** may be press-fitted into the sockets **9***a,* **9***b* from both sides in the direction of the axis D.

[0092] For example, the attachment pins **29** may be simple round bars and may be press-fitted from both sides in the direction of the axis D to a depth of about half that of the pin hole **9***h.*

[0093] In this embodiment, the connection structure **10** is attached between the sockets **9***a,* **9***b,* and the connection structure **10'** is attached between the sockets **9***a*', **9***b*'. Alternatively, the connection structure **10'** attached between the sockets **9***a*', **9***b*' at the other ends of the links **5***a,* **5***b* may be omitted.

[0094] In this case, the coil spring **31** of the connection structure **10** may have a spring constant that is sufficient to hold the four ball joints **7***a,* **7***b,* **7***a*', **7***b*'.

[0095] Although embodiments of the present disclosure have been described in detail above, the present disclosure is not limited to the individual embodiments described above. Various additions, substitutions, changes, partial omissions, and so forth may be made to the embodiments to the extent that the resulting embodiments do not depart from the gist of the present invention or from the idea and purpose of the present invention derived from the claims and equivalents thereto. For example, the order of individual operations and the order of individual processes in the above mentioned embodiments are illustrated merely as examples and are not limited to this.

1. A connection structure interconnecting two links, the two links having longitudinal axes and being spaced apart from each other in parallel and operated, the connection structure comprising:

bushings attached to the links so as to be rotatable about an axis perpendicular to a plane containing the two longitudinal axes;

a biasing mechanism spanning between the bushings of the two links and applying an elastic restoring force to the two bushings in a direction in which the two bushings move closer together; and

a connection member having two attachment holes through which the bushings can pass in a direction of the axis and regulating a distance between the bushings to be less than or equal to a prescribed distance,

wherein the bushings include claw portions projecting radially outward at part of, in a peripheral direction, an end portion of each of the bushings in the direction of the axis,

the attachment holes are formed in a shape allowing the bushings to pass therethrough only at a prescribed attachment phase around the axis coinciding with the claw portions of the bushings, and

a phase of the bushings around the axis is regulated by the elastic restoring force to a phase that does not match the attachment phase.

2. The connection structure according to claim **1**, wherein the bushings are disposed on both sides of the links in the direction of the axis,

the biasing mechanism is connected to the bushings on both sides of the links.

3. The connection structure according to claim **1**, wherein the biasing mechanism includes an elastic body that generates the elastic restoring force and hooks disposed at both ends of the elastic body and configured to hook onto the bushings.

4. The connection structure according to claim **3**, wherein the hooks and bushings have shapes that are maintained in a state where relative rotation around the axis is prohibited when the hooks are hooked.

5. The connection structure according to claim **4**, wherein at least one end of the elastic body and the hooks are connected so as to be rotatable relative to each other around an axis parallel to the axis.

6. The connection structure according to claim **5**, wherein a groove that is recessed radially inward and is configured to catch the hooks is provided at a midway position, in the direction of the axis, of a circumferential surface of each of the bushings.

7. A parallel link robot comprising:

a base;

a movable part disposed away from the base;

a plurality of arms pivotably connected to the base;

pairs of links having longitudinal axes parallel to each other that connect the arms to the movable part; and

the connection structure according to claim **1** provided between each of the pairs of links.

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